

JAKEB KNOWLES

Product-Focused Full-Stack Software Engineer | Product Builder

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Product-focused full-stack software engineer with 3+ years of experience turning ambiguous product problems into shipped SaaS, API, mobile, and real-time web applications. Strongest when working close to customers, product teams, and technical constraints at the same time: discovery, demos, architecture, frontend, backend, deployment readiness, and the details that decide whether a product survives real use.

CORE STRENGTHS

Product engineering: client discovery, MVP demos, requirements gathering, product trade-offs, technical leadership.

Full-stack delivery: React, TypeScript, AdonisJS, Laravel, PHP, Symfony/Slim, Node.js, REST APIs, React Native, Next.js.

Data and real-time systems: PostgreSQL, TimescaleDB, PostGIS, Aurora, MySQL, SQLite, SSE, Mapbox/WebGL, multi-tenant access control.

Engineering leverage: test strategy, CI quality gates, legacy modernisation, documentation systems, AI-assisted development workflows.

EXPERIENCE

Software Engineer / Technical Lead - Simpro Software

Jun 2025 - Present | Brisbane, QLD | Hybrid

- Led the technical rebuild of Simpro's fleet management platform, replacing a 13-year-old legacy solution with a modern real-time SPA for field service customers.
- Acted as sole technical lead on a small product team, personally designing and implementing roughly 95% of the application across React, TypeScript, AdonisJS, TimescaleDB, Aurora, and Mapbox.
- Partnered directly with PMs, designers, executives, and customers to demo MVPs, gather requirements, validate product direction, and help retain multiple six-figure ARR accounts.
- Increased committed tracker demand from roughly 5k to 8k through customer demos and product discovery, with the product live in beta and broader release scheduled after testing and production hardening.
- Architected an approved real-time tracking platform designed to support up to 20k trackers across four regions using TimescaleDB, PostGIS, SSE, and a Go ingestion layer.
- Built a high-performance WebGL/Mapbox vehicle map capable of rendering roughly 1,000 vehicles smoothly, using batched SSE updates to avoid frontend performance degradation.
- Improved fleet tracking from legacy 2-minute polling toward near-real-time location updates, with architecture capable of processing tracker pings at roughly 2-second intervals.
- Implemented multi-tenant access controls using JWT identity propagation, licence gating, company-scoped models, OAuth2 client credentials, and secure data access boundaries.
- Built reporting, service booking, geosite arrival/departure, technician schedule, behaviour event, and notification workflows across the frontend and backend.
- Maintained roughly 85% test coverage using Japa, Vitest, and Playwright, with CI quality gates blocking failed test builds before deployment.
- Owned production readiness for beta/UAT deployment, including deployment start scripts, migration safety checks, environment readiness, and rollback-aware release processes.
- Contributed heavily to TimescaleDB procurement and technical evaluation, comparing options including QuestDB and MongoDB against time-series, PostGIS, migration, and operational needs.
- Developed internal AI-assisted engineering workflows using repo documentation, specialised agents, skills, hooks, and workflow prompts to accelerate planning, implementation, and review.

Additional Simpro Contributions

- Contributed to Simpro's public partner API built on Symfony/Slim, adding endpoints, fixing production defects, and supporting OAuth2-authenticated integrations used by hundreds of partners.
- Resolved 50+ partner API forum support threads, helping unblock external developers and identify documentation/product gaps.
- Rebuilt the API documentation experience by extending a Redocly documentation site with custom search, reducing navigation from minutes of manual browsing to seconds of direct lookup.
- Helped resolve multiple P1 API incidents, including cache-related failures introduced during a Slim v4 upgrade.
- Led frontend modernisation of a React/Django data automation platform, migrating from CRA to Vite and upgrading MUI v4 to v6.
- Reduced known frontend vulnerabilities from roughly 800 to 8 and improved build times from roughly 30 seconds to 2-3 seconds.
- Created reusable base table and infinite-scroll table components while refactoring Cypress E2E tests to protect against regressions.
- Promoted from Junior Software Developer to Senior Associate Software Engineer to Software Engineer within nine months, driven by delivery speed, technical ownership, and ability to solve ambiguous product problems.
- Delivered internal engineering talks on PHP/MCP, JavaScript frameworks, developer tools, and agentic engineering workflows.

Software Engineer - FONSEKA Innovations

Apr 2024 - Jun 2025 | Brisbane, QLD

- Built and maintained full-stack business applications across Laravel, Vue, React, Next.js, React Native, PHP, JavaScript, TypeScript, SQL, and Tailwind CSS.
- Developed high-traffic Laravel applications for large userbases, using event-driven architecture, workflow optimisation, and test coverage targets around 90%.
- Led delivery of Fertility Quest, a React Native and Expo mobile app backed by a Laravel API, token-based authentication, user profiles, activity logging, Stripe subscription tiers, quizzes, and structured course content.
- Delivered the Fertility Quest MVP in one month against an initial two-month estimate, balancing mobile UX, backend API design, subscription handling, and release readiness.
- Built an internal project management platform spanning lead generation, quoting, order handling, MSA workflows, dashboards, and sales lifecycle metrics.
- Used TDD across Laravel and React with PHPUnit, Pest, and Jest to improve confidence across core business workflows.
- Rebuilt core CRUD and document-generation workflows in a 10+ year-old service platform using Vue and Laravel, integrating third-party services including Google Maps.
- Improved maintainability across five large-scale applications by increasing test coverage and reducing high-risk change patterns in legacy codebases.

Product Builder / Indie Developer - JK Web Dev

Jul 2023 - Present | Side projects and independent builds

- Build and ship independent web and mobile products across Laravel, React, React Native, Expo, Angular, TypeScript, Stripe, SQLite, MySQL, GitHub Actions, and Linux hosting.
- **Trendsetter:** Building an AI-powered goal-setting app that turns natural-language conversations into structured plans, timelines, calendar events, progress logs, and point-based goal tracking. Built with Expo, React Native, Laravel, SQLite, Zustand, NativeWind, OpenAI API, Pest, Jest, GitHub Actions, and GitHub Projects.
- **Picklewear:** Designed and built a custom Laravel and React e-commerce platform for an online pickleball retailer, including admin dashboards, product management, order management, Stripe payment flows, AusPost order tracking, and roughly 95% backend test coverage.

EDUCATION AND CERTIFICATIONS

Bachelor of Information Technology, Major in Software Development - Griffith University, Oct 2023

Academic Improvement Award Recipient, 2023 | Fundamentals of Digital Marketing - Google, Mar 2024